CAD/CAM

III B. Tech. I Semester: MECHANICAL ENGINEERING								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME25	PCC	L	T	P	C	CIE	SEE	Total
		2	0	0	2	40	60	100

COURSE OVERVIEW:

This course in CAD/CAM provides a comprehensive exploration of the fundamentals of CAD, CAM, and Automation, covering topics such as geometric modelling, NC control, computer-aided process planning, computer-aided manufacturing, computer-aided quality control, and Computer Integrated Manufacturing (CIM), with a focus on the benefits of utilizing these technologies in the product design and development cycle.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Describe basic concepts of computer hard ware and software connected to CAD & CAM.
2. Explain wire frame modelling, surface modelling and solid modelling.
3. Describe about NC, CNC, DNC, adaptive control systems, Group Technology and can develop NC part programs.
4. Describe computer aided process planning and manufacturing resource planning.
5. Explain concepts of computer aided quality control and computer integrated manufacturing.

UNIT-I INTRODUCTION

Fundamentals of CAD, CAM, Automation, design process, Application of computers for design, Benefits of CAD, Computer peripherals for CAD, CAD software- definition of system software and application software, CAD database and structure. Product development cycle.

UNIT-II GEOMETRIC MODELING

Introduction to Geometric Modelling, **Wire frame modelling:** Definition, advantages, dis-advantages, wire frame entities. **Surface modelling:** Definition, advantages, dis-advantages, surface entities. **Solid Modelling:** Definition, constructive solid geometry, advantages.
Curve representation: Implicit and explicit forms of straight line, circle. **Parametric representation of synthetic curves:** Cubic Spline, Bezier Curve, B Spline Curve.

UNIT-III NC CONTROL & GROUP TECHNOLOGY

NC Control: Introduction, Elements of NC system, NC part programming: Methods of NC part programming, manual part programming, Computer assisted part programming, CNC, DNC, and Adaptive Control Systems.
Group Technology: Part families, Parts classification & coding. Production flow analysis, Machine cell design using rank order clustering technique.

UNIT-IV COMPUTER AIDED PROCESS PLANNING & COMPUTER AIDED MANUFATURING

Computer aided process planning: Difficulties in traditional process planning, Computer aided process planning: Retrieval type and Generative type.
Computer aided manufacturing: Material Resource Planning, inputs to MRP, MRP output records, Benefits of MRP, Enterprise Resource Planning, Capacity Requirements Planning

UNIT-V	COMPUTER AIDED QUALITY CONTROL AND CIM

Computer aided quality control: Automated inspection – Off-line, On-line, contact, Non-contact; Coordinate measuring machines, Machine vision, integration of CAQC with CAD/CAM.

Computer Integrated Manufacturing: Types of systems in CIM, CIM advantages and disadvantages.

Text Books:

1. P.N. Rao, CAD/CAM Principles and Applications, 3rd Edition, Tata McGraw Hill Education Pvt. Ltd., New Delhi, 2017.
2. Mikell P. Groover, Emory W. Zimmers, Jr., CAD/CAM Computer-Aided Design and Manufacturing, 1st Edition, Pearson Education, 2003.

Reference Books:

1. Ibrahim Zeid, R. Sivasubramanian, CAD/CAM Theory and Practice, 2nd Edition, McGraw Hill Education, 2009.
2. P. Radhakrishnan, S. Subramanian and V. Raju, CAD/CAM/CIM, 4th Edition, New Age International Publishers, 2018.
3. Chennakesava R. Alavala, CAD/CAM Concepts and Applications, Prentice Hall India Learning Pvt. Ltd., 2008.
4. Farid Amirouche, Principles of Computer Aided Design and Manufacturing, 2nd Edition, Pearson, 2004.
5. Warren S Seames, Computer Numerical Control Concepts and programming, 4th Edition, Thomson Learning, 2022.